tokentax: Measuring the Tokenizer Cost Penalty Across 203 Languages

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Abstract

The same sentence costs 1× tokens in English but often 5–12× in Shan, Santali, or Dzongkha under current LLM tokenizers, because tokenizer vocabularies are trained on English-heavy corpora. Petrov et al. [2023] and Ahia et al. [2023] established this "token tax" for 2023-era tokenizers, but no maintained, versioned resource reports current numbers for the tokenizers actually in production use today. We present `tokentax`, a measurement suite, versioned results dataset, and library that computes the token premium, and its 95% confidence interval via `evalci` [Chandrahas, 2026], for 8 tokenizers (GPT-4o, GPT-4, GPT-2, Llama 3, Qwen2.5, DeepSeek-V3, Mistral, Gemma-2) across all 203 non-English FLORES-200 languages [NLLB Team et al., 2022]. The pipeline reproduces Petrov et al.'s published GPT-2/GPT-4 premium ratios within 1.1% relative error on five languages, a direct validation against prior work rather than only internal self-consistency. Across the full sweep, mean premium ranges from 2.13× (Gemma-2) to 4.48× (GPT-2) – tokenizers released in 2024 uniformly show a lower mean premium than GPT-2 (2019) and GPT-4's cl100k_base (2022), consistent with vocabulary enlargement and more diverse pretraining data, though vocabulary size and training-corpus composition are confounded in this comparison and we do not claim to isolate their individual contribution. A domain-robustness check against a second, non-encyclopedic corpus (mixed-domain OPUS-100 text and/or a Bible-translation corpus) for the 30 highest-premium languages finds the premium replicates within a median 14% across 264 (language, tokenizer, domain) comparisons where a second corpus exists at all – but 9 of those 30 languages have no modern, ungated parallel corpus available on Hugging Face for a second domain, a scarcity that is itself part of the token-tax story. `tokentax` is available at `https://pypi.org/project/tokentax/` (source: `https://github.com/Shreyaskc/token-tax`), with the full results dataset and an interactive explorer at `https://huggingface.co/datasets/shreyaskc/tokentax-results-v1` and `https://huggingface.co/spaces/shreyaskc/tokentax-explorer`.

1 Introduction

Token-based API pricing and fixed context windows mean tokenizer inefficiency is a real economic and capability penalty, not a cosmetic one. If a language costs $k\times$ as many tokens as English for the same content, its speakers pay $k\times$ as much per query, retain $1/k$ as many effective few-shot examples in a fixed context window, and hit context truncation $k\times$ sooner. Petrov et al. [2023] first quantified this systematically, finding tokenization-length disparities of up to 15× across languages for tokenizers available in 2023; Ahia et al. [2023] independently showed that OpenAI's commercial API pricing, applied uniformly per token, overcharges speakers of many supported languages for equivalent utility. Rust et al. [2021] showed the same underlying disparity degrades downstream monolingual task performance, not just cost.

These are foundational results, but the tokenizers they measured are no longer the ones in production use. GPT-2's tokenizer (2019) and cl100k_base (2022, GPT-3.5/GPT-4) have both been superseded by larger, more multilingual-aware vocabularies – OpenAI's o200k_base (GPT-4o), and open-weight vocabularies from Meta, Alibaba, DeepSeek, Mistral, and Google. Whether the token tax has actually improved with these newer vocabularies, and by how much, is a question every multilingual-fairness paper since 2023 has had to re-derive ad hoc, because no maintained, versioned, statistically rigorous resource reports current numbers. This is precisely the gap a living benchmark should fill: the numbers change with every tokenizer release, which argues for a re-runnable measurement suite and a versioned dataset over a one-off paper.

We present `tokentax`: (1) a small Python library that computes the token premium, bytes/characters-per-token, and effective context-window shrinkage for any tokenizer against any parallel corpus, with confidence intervals via `evalci` [Chandrahas, 2026]; (2) a full sweep of 8 current tokenizers against all 203 non-English FLORES-200 languages, validated by reproducing Petrov et al.'s published numbers; (3) a domain-robustness check against a second corpus for the 30 most heavily taxed languages; and (4) a versioned results dataset and interactive explorer, so the resource stays current as new tokenizers ship.

2 Related Work

Petrov et al. [2023] is the direct precedent for this work: a systematic study of tokenization-length disparity across languages for GPT-2, GPT-3, GPT-4, and several open tokenizers of the time, using FLORES-200 as the parallel corpus – the same choice we make, for the same reason (Section 3). Ahia et al. [2023] took the complementary economic angle, translating token counts into actual API dollar costs and utility-per-dollar across 22 languages, and is the direct precedent for our cost-analysis framing (Section 5.5). Rust et al. [2021] showed that a designated monolingual tokenizer, not just pretraining data volume, materially affects downstream task performance – the capability-side consequence of the same disparity we measure on the cost side. NLLB Team et al. [2022] introduced FLORES-200, the parallel-sentence corpus this entire line of work, including ours, depends on.

`tokentax`'s contribution relative to this line of work is not a new finding about *why* the disparity exists – that question is settled – but a maintained instrument for *how much* it currently is, built to be re-run as tokenizers change, with every reported number carrying a confidence interval rather than a bare point estimate, and with an explicit domain-robustness check rather than a single-corpus measurement.

3 The `tokentax` Suite

Premium ratio. For a tokenizer T and an aligned sentence pair (s_{en}, s_ℓ) expressing the same content in English and language ℓ, the premium ratio is $|T(s_\ell)|/|T(s_{\text{en}})|$, where $|T(\cdot)|$ is token count. We compute this only on *aligned parallel* sentences, never independent monolingual corpora: only a shared-meaning pair makes "language ℓ costs $k\times$ as many tokens as English for the same content" a defensible claim, since monolingual corpora in different languages differ in register, sentence length, and content independent of tokenization efficiency.

Corpus. We use FLORES-200 devtest [NLLB Team et al., 2022]: 1,012 sentences, professionally translated in parallel into 200+ languages, covering 203 languages with an English pairing in our sweep. FLORES-200 is gated on Hugging Face (a free, instant license click, not a manual review).

Tokenizers. We study 8 tokenizers: OpenAI's o200k_base (GPT-4o) and cl100k_base (GPT-4), and the legacy GPT-2 vocabulary (kept only for the validation in Section 4); Meta's Llama 3; Alibaba's Qwen2.5; DeepSeek-V3; Mistral (v0.3); and Google's Gemma-2. Two deviations from the obvious choice, both discovered by actually running the pipeline rather than assumed in advance: `meta-llama/Meta-Llama-3-8B` requires Meta's manual license approval rather than an instant click-through, so our Llama 3 tokenizer resolves to the widely-used, ungated `NousResearch/Meta-Llama-3-8B` mirror of the identical vocabulary; and Mistral's tokenizer turned out not to be gated at all once tested, only missing a `protobuf` runtime dependency. Anthropic's Claude is excluded: it has no downloadable tokenizer artifact, only a token-counting API endpoint, which would require every user reproducing this dataset to hold an API key. Gemma-2's SentencePiece vocabulary is also the closest available open proxy for Gemini's tokenizer, which likewise has no public artifact; we report it only under the Gemma-2 name and do not present it as a Gemini measurement.

Confidence intervals. Every reported premium ratio is a percentile bootstrap 95% confidence interval over the 1,012 per-sentence ratios, computed by `evalci.ci(method="bootstrap")` [Chandrahas, 2026] – not reimplemented, per this project's cross-artifact reuse policy.

Implementation. `tokentax` is a pure-Python library (`numpy`, `pandas` [Harris et al., 2020, McKinney, 2010]), loading tokenizers via `tiktoken` for OpenAI vocabularies and Hugging Face `transformers` [Wolf et al., 2020] for every open-weight one.

4 Validation Against Prior Work

Rather than validate only against our own pipeline's internal consistency, we reproduce Petrov et al.'s own published per-language token totals over FLORES-200 devtest for GPT-2 and GPT-4's cl100k_base, for five languages spanning a range of scripts and premium magnitudes.[1] Table 1 shows agreement within 1.1% relative error for every (language, tokenizer) pair – well inside the 2% target we set before running it, and consistent with the small differences expected from FLORES-200 dataset revisions since 2023.

Language	Tokenizer	Published	Reproduced	Rel. error
Amharic	GPT-2	7.787	7.702	1.1%
Amharic	GPT-4	7.678	7.593	1.1%
Tamil	GPT-2	15.582	15.537	0.3%
Tamil	GPT-4	7.651	7.644	0.1%
Burmese	GPT-2	16.891	16.813	0.5%
Burmese	GPT-4	11.702	11.618	0.7%
Hindi	GPT-2	7.459	7.413	0.6%
Hindi	GPT-4	4.794	4.759	0.7%
Standard Arabic	GPT-2	4.400	4.358	1.0%
Standard Arabic	GPT-4	3.037	3.027	0.3%

Table 1: Reproduction of Petrov et al. [2023]'s published premium ratios. Maximum relative error 1.1% across 10 (language, tokenizer) pairs.

[1] Reference values computed from `https://github.com/AleksandarPetrov/tokenization-fairness`, the paper's own released data, as a ratio of total tokens over the full devtest set (matching their methodology), not a mean of per-sentence ratios.

5 Results

5.1 Distribution

Across 8 tokenizers × 203 languages (1,624 (tokenizer, language) pairs), the mean premium is 2.96×, the median 2.14×, and the range spans 0.94× to 19.12×. The single sub-1× value is Mandarin Chinese under DeepSeek-V3's tokenizer (0.94×) – a genuine exception, not noise (the 95% CI excludes 1), plausibly because a vocabulary trained with substantial Chinese-language data can represent common multi-character sequences more compactly per token than English byte-pair merges do.

5.2 Which languages are taxed the most

Figure 1 shows the 15 languages with the highest mean premium across all 8 tokenizers. The pattern is stark: every language above 6.5× uses a script with a small digital footprint relative to its speaker population – Shan (Myanmar script variant), Santali (Ol Chiki), Dzongkha and Tibetan (Tibetan script), Tamasheq and Tamazight (Tifinagh), Lao, Khmer, and several South Asian scripts (Odia, Malayalam, Sinhala, Kannada, Telugu). At the opposite end, the 10 lowest-premium languages are mostly Latin-script European languages (German, Dutch, Swedish, Norwegian Bokmål, Galician, Spanish, Portuguese) alongside Indonesian, Pangasinan, and Simplified Chinese, ranging 1.48–1.58× – close to, but still measurably above, English's own baseline.

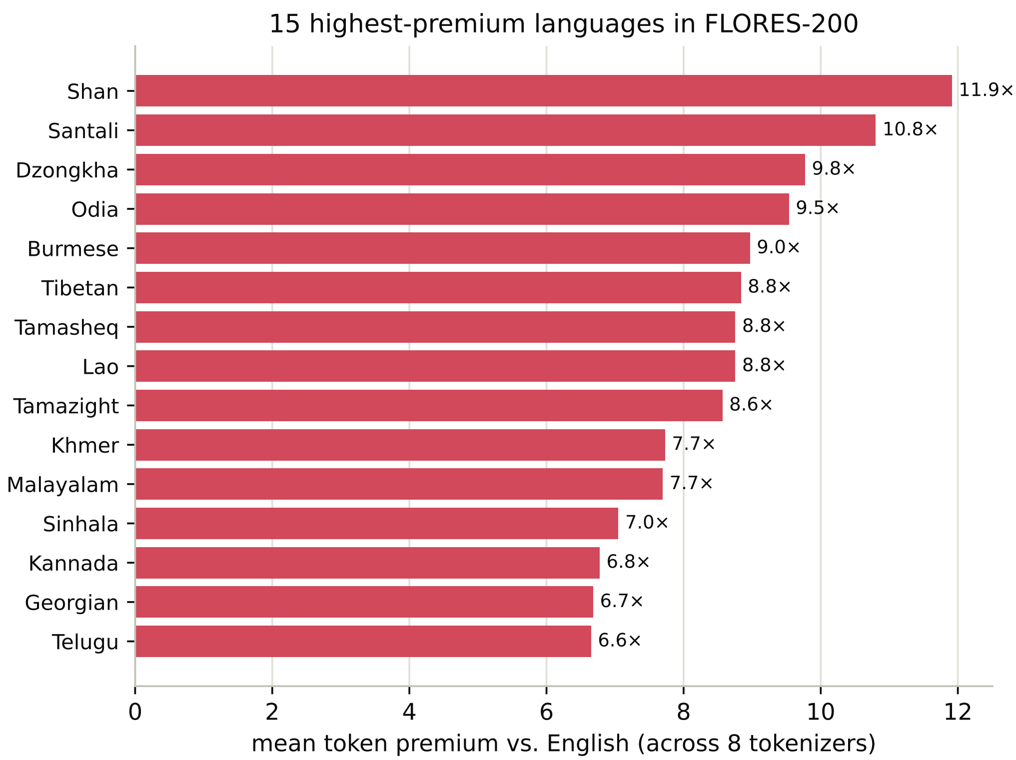

Figure 1: 15 languages with the highest mean token premium vs. English, averaged across all 8 tokenizers in the sweep.

5.3 Newer tokenizers, lower premium

Figure 2 orders the 8 tokenizers by mean premium. The four tokenizers released in 2024 with the lowest mean premium (Gemma-2, GPT-4o, DeepSeek-V3, Qwen2.5, all $\leq 2.8\times$) sit uniformly below GPT-4's 2022 cl100k_base ($3.42\times$) and GPT-2's 2019 vocabulary ($4.48\times$). This is consistent with larger vocabularies and more linguistically diverse pretraining corpora across successive tokenizer generations, but release year, vocabulary size, and training-corpus composition are confounded in a sample of 8 tokenizers, and we do not claim to have isolated which factor drives the trend – only that it holds directionally across every 2024-era tokenizer we measured against both older baselines.

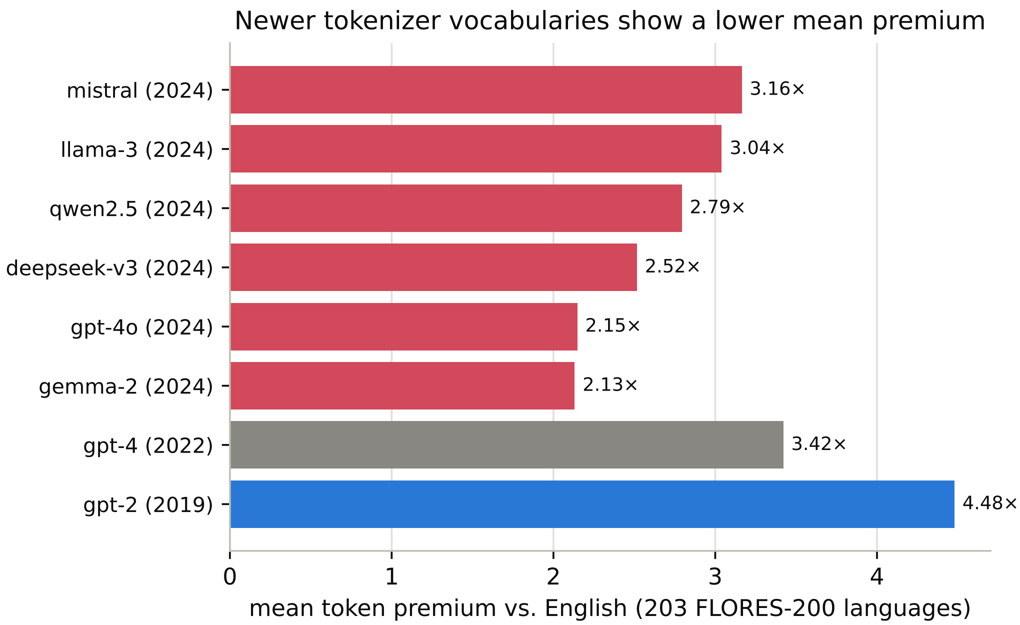

Figure 2: Mean token premium by tokenizer, colored by approximate release era of the associated model family (blue: 2019, gray: 2022, red: 2024). Release years reflect the model family's public release, not necessarily the exact date the tokenizer vocabulary itself was frozen.

5.4 Domain robustness

FLORES-200 is encyclopedic/news-register text; a premium ratio that only replicates on that one register is a weaker claim than one that holds across registers. For the 30 languages with the highest mean FLORES premium, we additionally measured the premium on a second corpus where one exists: mixed-domain OPUS-100 text and/or a Bible-translation corpus (religious register), each with its own aligned English pairing (not FLORES's).

Of these 30 languages, 21 have at least one second-domain corpus available; the other 9 – Shan, Santali, Tibetan, Tamasheq, Tamazight, Lao, Tigrinya, Manipuri, and Kabiyè – have no modern, ungated, actively-maintained parallel corpus on Hugging Face for a second domain at all. We report this gap explicitly rather than silently narrowing the check to the languages that happen to have data: the languages with the highest token tax are, by this evidence, also among those with the least available parallel data to independently corroborate it, which is itself a relevant fact for multilingual-NLP resource allocation.

Across the 264 (language, tokenizer, domain) comparisons where a second corpus does exist, the median relative difference from the FLORES estimate is 14% (mean 17%) – the premium mostly replicates across register. 23 of 264 comparisons (8.7%) diverge by more than 35%, concentrated in a handful of (language, domain) combinations rather than spread evenly: Kannada on OPUS-100 (up to 107% divergence, likely a short, UI-string-heavy test split rather than a general finding), and Uyghur and Sanskrit on the Bible corpus (44–72% divergence, plausibly small and register-idiosyncratic samples in those specific files). We report these as corpus-specific caveats on those particular language/domain pairs, not as evidence against the premium ratio generally.

5.5 Cost analysis

Following Ahia et al. [2023]'s framing, Table 2 converts premium ratios into dollar terms: at GPT-4o's published input-token price ($2.50 per million tokens, a snapshot we mark unverified in our released pricing table and re-check before every citation of a dollar figure), the same 1-million-token quantity of content costs $2.50 in English but up to $34.82 in Santali – a 13.9× markup for identical meaning, before any consideration of the additional latency and context-window cost of the extra tokens.

Language	Premium (GPT-4o)	Cost per 1M-token-equivalent
Santali	13.93×	$34.82
Tamasheq	10.41×	$26.04
Tamazight	10.26×	$25.65
Dzongkha	9.22×	$23.06
Lao	8.38×	$20.94
Tibetan	8.23×	$20.56
Shan	8.10×	$20.26
Tigrinya	6.05×	$15.13
Amharic	5.85×	$14.63
Odia	5.05×	$12.62
English (baseline)	1.00×	$2.50

Table 2: Cost to process 1 million English-tokens' worth of equivalent content, by language, at GPT-4o's published input-token price. Pricing is an unverified snapshot; see Limitations.

6 Limitations

Pricing figures (Section 5.5) are a static, manually-maintained snapshot explicitly marked `verified: false` in the released pricing table; the library warns on load until a maintainer confirms current provider pricing, and dollar figures should be treated as illustrative of magnitude, not as a citable current price. Claude is excluded from every result in this paper for the reason given in Section 3, not because it is exempt from the phenomenon. The Llama 3 tokenizer is measured via an ungated third-party mirror, not Meta's own gated repository, though we have no reason to expect the vocabulary differs. Gemma-2 is reported only as itself, not as a validated proxy for Gemini's (unpublished) tokenizer. The domain-robustness check (Section 5.4) covers 21 of the 30 highest-premium languages, not all 203, and the 9 uncovered languages are systematically the most extreme cases, not a random sample – so the 14% median replication figure should not be extrapolated to languages outside the 21 actually checked. FLORES-200 devtest is a single snapshot of encyclopedic/news-register text; even where a second domain is available, both corpora are translations rather than natively-authored text, which may itself introduce a shared translationese effect that a purely monolingual comparison would not have. Finally, every number in this paper reflects the specific tokenizer revisions pinned at measurement time (Section 8); tokenizer vocabularies are periodically updated by their maintainers, and the versioned dataset, not this PDF, is the source of truth for current numbers.

7 Ethics Statement

This work is a measurement, not a system: it does not collect, generate, or redistribute personal data, and the released dataset contains only token counts, premium ratios, and confidence intervals – never the underlying FLORES-200, OPUS-100, or Bible-corpus sentence text, which remain under their original sources' own licenses. The motivating concern is fairness: speakers of the languages in Table 2 and Figure 1 pay materially more for equivalent access to commercial language models, a disparity this paper aims to make legible and citable rather than to newly create. We are not aware of a plausible dual-use or harm pathway specific to this artifact; the intended use is informing tokenizer design, pricing-policy discussion, and multilingual-NLP methods sections that need a current, citable number.

8 Availability

`tokentax` is released under the MIT license (results dataset under CC0) and available via `pip install tokentax` (`https://pypi.org/project/tokentax/`). Source, the full sweep and domain-robustness scripts, and tests are at `https://github.com/Shreyaskc/token-tax`. The versioned results dataset – 1.64M raw per-sentence rows, the CI-backed summary table, and the domain-robustness comparisons – is at `https://huggingface.co/datasets/shreyaskc/tokentax-results-v1`, and an interactive heatmap and cost calculator at `https://huggingface.co/spaces/shreyaskc/tokentax-explorer`. Continuous integration runs the test suite across Python 3.9–3.12 on every push.

9 Conclusion

The token tax is not a 2023 finding that has since been fixed: it persists in every tokenizer we measured, ranging from 1.48× to over 19×, and even its most improved 2024-era vocabularies still charge Georgian, Kannada, or Malayalam speakers roughly 2–3× English's rate for identical content. What has changed is which vocabulary a researcher or policymaker should cite a number from – and that number will keep changing. `tokentax` is built to be re-run, not re-derived: the measurement suite, the versioned dataset, and the explorer are designed so that the next tokenizer release gets a new, citable number within days, not another ad hoc re-implementation of Petrov et al.'s original analysis.

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